Key features of hyperbola

1

a 0

b ∞

c $x = 0$

d $y = 0$

e $y = x, y = -x$

2

a -4

b Negative

c Positive

d 2 and 4

3

a 0

b $-\infty$

c ∞

d 0

e 0

4

a

x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y	-1	-2	-4	-20	-200	200	20	4	2

b 0

c $x = \frac{2}{y}$

d 0

e As x approaches 0 from the right, the function value approaches ∞. As x approaches 0 from the left, the function value approaches $-\infty$.

As x approaches ∞ and $-\infty$, y approaches 0. This is called the limiting value of the function.

5 6

6

a Yes

b No

c No

7

a $x = 0$

b $y = 0$

c All real values of x, except $x = 0$.

d All real values of y, except $y = 0$.

8 The function will be undefined.

9

a $x = 0$ only

b All real values of x, except $x = 0$

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c $y = 0$

d All real values of y, except $y = 0$.

10 Looking at $y = \frac{2}{x}$, notice that the numerator 2 is non-zero, and so $\frac{2}{x}$ can never be equal to 0.

Another approach is to rearrange the equation and make x the subject. Then we get $x = \frac{2}{y}$.

In this form we can see that the denominator y cannot be equal to 0.

11

a 2

b 1

c $y = \frac{4}{x}$

d The graph moves further from the axes

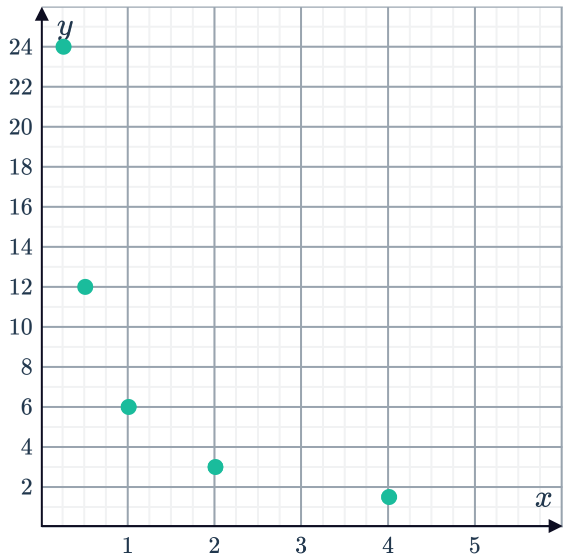
Graphs of hyperbolas

12

a

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y	24	12	6	3	$\frac{3}{2}$

b



13 She can label a point on the graph.

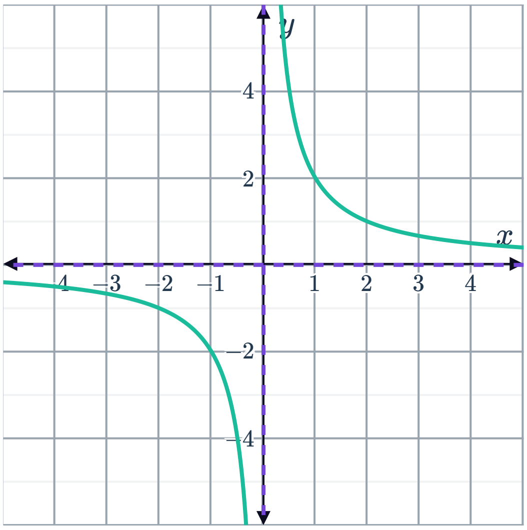
14

a

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	-1	-2	-4	4	2	1

ii



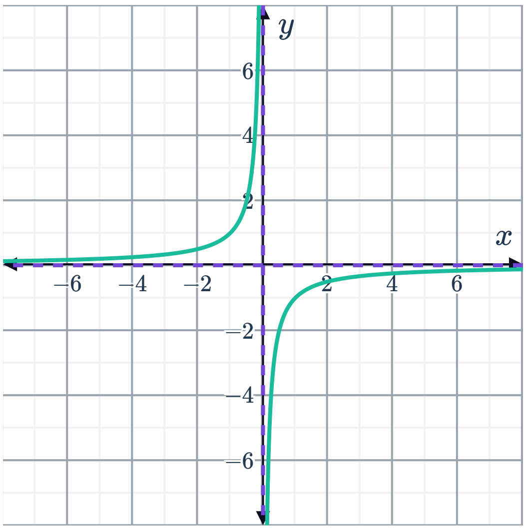
iii 1 and 3

b

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{2}$	1	2	-2	-1	$-\frac{1}{2}$

ii



iii 2 and 4

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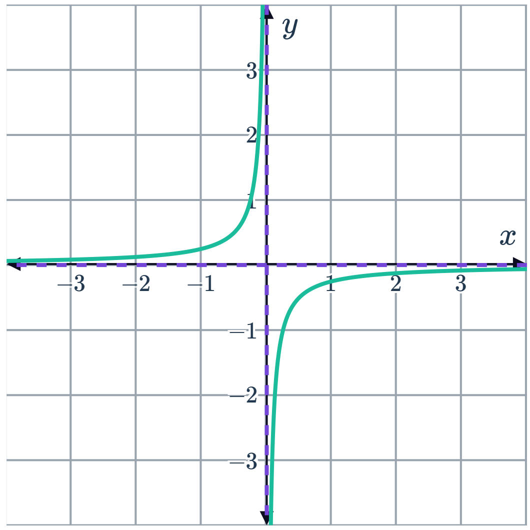
c

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{4}$	$-\frac{1}{8}$

ii

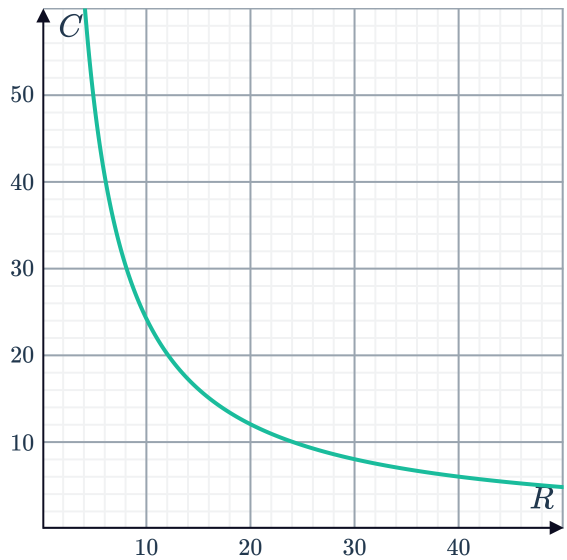


iii 2 and 4

Applications

15

a



b The current decreases.

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16

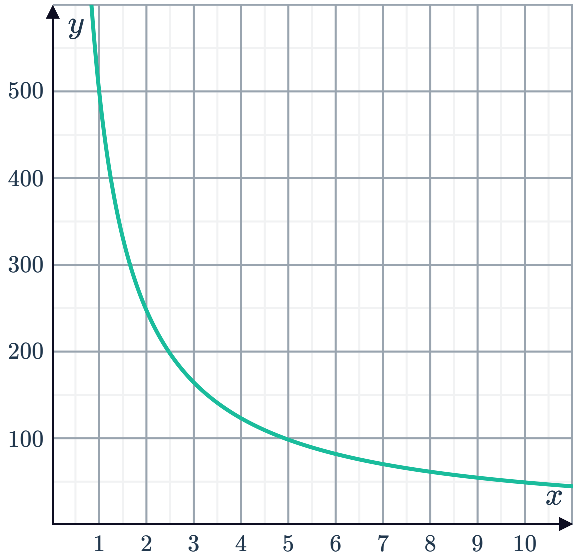
a $\$245.00$

b $y = \dfrac{490}{x}$

c

x	1	2	3	4	5	6
y	490	245	163.33	122.5	98	81.67

d



e Inverse variation

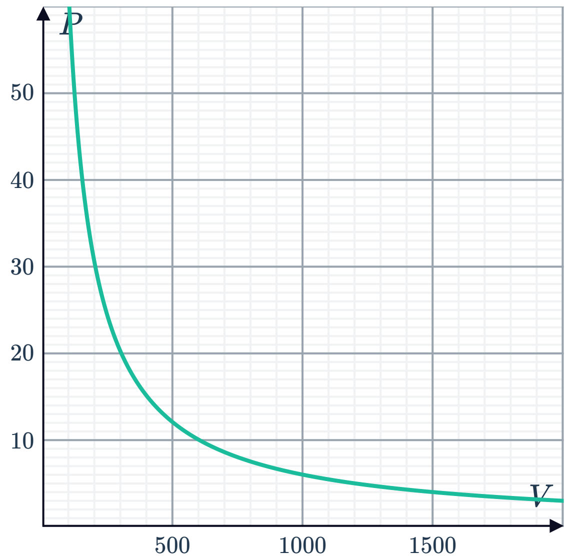
17

a -2

b 32 units^2

18

a



b 6000 kg/cm^2

c The pressure decreases.

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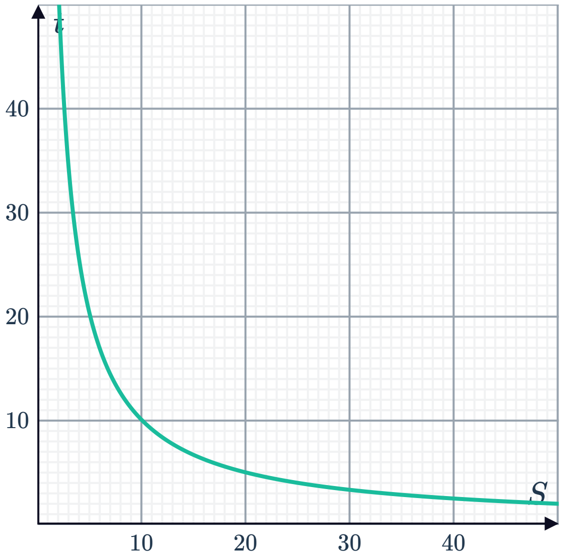
19

a



b 10 hours

c 2 hours

d The speed must increase.